Alexis Valencia

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EDUCATION

Carnegie Mellon University (CMU) **Pittsburgh, PA**
Master of Science in Computational Biomedical Engineering – Research | GPA: 3.67/4.0 May 2026
Thesis: A Fully Flexible Wearable Continuous-wave Near-infrared Spectroscopy for Continuous Cerebral Hemodynamic Monitoring
Relevant Coursework: Brain Computer Interface, Neural Signal Processing, Fundamentals to Computational BME

University of California – Riverside (UCR) **Riverside, CA**
Bachelor of Science in Bioengineering | GPA: 3.58/4.0 June 2024
Relevant Coursework: Biosystems and Signal Analysis, Systems Neuroscience

SKILLS

Programming & Software: C/C++, MATLAB, Python, C#, JavaScript, GitHub, Fusion 360 CAD, Unity (XR/AR/VR)
Firmware & Embedded Systems: Cortex-M (STM32), Arduino, Communication Protocols (SPI, UART, I2C, BLE, DMA), ADC Sampling, Real-Time Data Acquisition, Peripheral Interfacing, Analog Front-End (AFE) optimization
Hardware & Analog/Optical Sensing: Flexible PCB Design (Altium Designer), Oscilloscope, Logic Analyzer, Soldering, Rapid Prototyping, Multimeter Troubleshooting
Signal Processing & Biomedical Systems: Time-Series Analysis, Feature Extraction, Classification, Signal Acquisition

RESEARCH EXPERIENCE

Research Associate | Pittsburgh, PA July 2026 – December 2026
Graduate Research Assistant July 2024 – May 2026
Wearable Near-Infrared Spectroscopy **|** Biophotonics Lab at CMU

- Refine STM32 firmware and **BLE** wireless streaming, advancing the wearable toward field deployment
- Designed and prototyped Neuro-FLux, a fully flexible wearable CW-NIRS brain monitor, achieving **36.14 dB SNR** versus 34.49 dB for a commercial system
- Developed STM32 firmware in C (SPI + DMA, I2C, UART), sustaining stable **100 Hz** real-time optical acquisition
- Validated timing with an oscilloscope and logic analyzer; improved SNR **~2 dB** via sensor interfacing and AFE tuning
- Designed a 4-layer flex PCB in Altium, selecting photodetector and AFE for a **56.6 × 24.4 mm** conformable sensor
- Integrated sensors, ADC sampling, and peripheral communication on the MCU for reliable real-time acquisition

Undergraduate Research Assistant | Riverside, CA June 2023 – June 2024
Augmented Reality Experiment for Schizophrenia Study **|** Systems Neural Engineering Laboratory at UCR

- Led development of an **AR experimental platform** on **Magic Leap 2** for schizophrenia behavioral studies
- Built **Unity (C#)** environments with hand tracking so participants completed tasks with natural movements

PROFESSIONAL EXPERIENCE

AR/VR Developer | XCITE Center for Teaching and Learning | Riverside, CA March 2023 – June 2024

- Developed interactive XR lessons in **Unity (C#)** for Meta Quest and mobile, making course concepts hands-on
- Led a team of **three** building an encryption lesson, demoed live to **60+ students** and refined from feedback

User Services Intern | TAMIU OIT | Laredo, TX October 2020 – August 2021

- Reimaged new computers to school-system standards and connected them to the campus network
- Resolved faculty and staff technology work orders by troubleshooting hardware and software issues

3D Printer Specialist | Clever Minds Tutoring Center | Laredo, TX June 2020 – August 2023

- Troubleshot print failures, material issues, and hardware inconsistencies, improving fabrication reliability
- Trained **three** staff on the full 3D printing workflow, from online CAD models to printed toys for children
- Gave **~20** children hands-on 3D printing experiences, supporting the center's STEM education initiatives

Game Room Team Leader | Peter Piper Pizza | Laredo, TX November 2018 – March 2020

- Led training and task delegation for **five** employees, keeping the game room clean and machines functional
- Managed weekly budgeting by collecting game profits and allocating funds to restock ticket prizes
- Maintained **40+ games** by replacing mechanical parts and reprogramming software to keep machines running

SELECTED PROJECTS

Improving 2-D Motor Imagery BCI Classification | Course Project, CMU October 2025 – December 2025

- Conducted EEG motor-imagery experiments and analyzed electrophysiology data for machine learning classification
- Built a Python pipeline using **bandpass filtering** and **Common Spatial Patterns (CSP)** feature extraction
- Trained a **Random Forest classifier**, improving F1-score from **50.9% to 78.8%** over a baseline classifier

Optical Coherence Tomography and Vibrometry Endoscope | Capstone Project, UCR September 2023 – June 2024

- Co-designed a **compact OCT-based diagnostic device** for hearing loss, winning NIH DEBUT Challenge **2nd Prize**
- Engineered **seven** SLA-printed optomechanical mounts with micro-adjustment, holding ±50 µm tolerance
- Aligned the optical path to **49.9%** throughput while reducing weight and keeping return power consistent